

Probability and Statistics. Week 2

1. Prove that if $B \subset A_1 \cup \dots \cup A_n$ where $A_i \cap A_j = \emptyset$ for $i \neq j$, then $P(A_1|B) + P(A_2|B) + \dots + P(A_n|B) = 1$.
2. Two different numbers are chosen at random from the set $\{1, 2, \dots, 100\}$. What is the probability that the first one is greater than the second?
3. There were 26 white balls and 14 black balls in the urn. One of the balls has disappeared. After that one ball is taken at random from the urn. Determine the probability that the ball is black.
4. A coin is flipped until one does not get a sequence "heads-tails-heads". What is the probability that the coin has been flipped
 - (a) exactly 6 times;
 - (b) exactly 7 times.
5. A card is drawn at random from a deck of 36 cards. Let A = the card is a jack and B = the card has a black suit. Find the probability of $A \cup B$.
6. Three marksmen are shooting at a target, their probabilities of hitting the target are equal to 0.5, 0.4 and 0.3 respectively (the results are independent). Each of them makes exactly one shot. What is the probability that:
 - (a) The target has not been hit;
 - (b) The target has been hit exactly two times;
 - (c) The target has been hit at least once.
7. It's been observed that 70% of those who attend lecture, get an A in Probability and Statistics. However, attendance is only at 35% for the semester. The probability of getting an A in Prob. and Statistics is $1/4$. Vseznajka got an A in Prob. and Statistics. What are the odds that he was attending lecture?
8. Ten people enter a ground floor of an eight-storey building. Find the probability that:
 - (a) the lift stops neither on the fourth nor on the fifth floor;
 - (b) the lift stops on the fourth floor and does not stop on the fifth;
 - (c) the lift stops both on the fourth and on the fifth floor;
 - (d) the lift stops at least twice;
 - (e) the lift stops at least twice, including the stop on the fourth floor.

Note: We use the following conventions for problems with lifts if it is not stated otherwise. 1) The floors are counted this way: ground floor, first floor, second floor, etc. 2) We assume that all people who have entered the lift at the ground floor can go out of it on each floor starting with the first one with equal probabilities and independently from the others. 3) No other people enter the lift as it goes up.

9. A woman arrives at the airport between 11 and 12 o'clock, waits for her husband for 10 minutes, and if he does not arrive, she hails a taxi and leaves the airport alone (and the husband has a lot of trouble!). Her husband shows up at the airport at some time between 11 and 11:40. He waits for 20 minutes, and if he does not meet his wife, he goes home (and he is in a lot of trouble!) What is the probability that the husband ends up in having lots of trouble?

10. Prove that
 - (a) $P(AB) \geq P(A) + P(B) - 1$;
 - (b) $P(A_1 A_2 \dots A_n) \geq P(A_1) + P(A_2) + \dots + P(A_n) - (n - 1)$.
11. Two players are taking turns in drawing cards from a deck of 52 cards. The first player starts with drawing one random card. If it is an ace, he wins the game. Otherwise he returns the card back to the deck, shuffles it properly, and then the second player makes his turn. The game continues until the winner is determined (the first one to draw an ace). Is this a fair game (i.e. the probability for each of the players to win are equal)? Does the game become more fair if the winner is the first to draw the ace of clubs?
12. There are 5 white and 3 black balls in the first urn, 2 white and 6 black balls in the second urn. Two random balls are transferred from the first urn to the second, and after that some ball is taken from the second urn. This ball happens to be white. Find the probability that two balls of different colours were moved from the first urn to the second.
13. There were 12 white and 8 black balls in the first urn, 8 white and 4 black balls in the second urn, 10 white balls in the third urn. The fourth urn, that had been empty, was filled with the balls: 6 balls from the first urn, 5 balls from the second, and 4 balls from the third (all the balls were chosen at random). After that two balls were taken out of the fourth urn, and they happened to be white. What is the probability that these balls originate from different urns?
14. All 100 tickets for the movie were bought up. A second before the hall opened, a mad granny burst into it and sat down in a random seat (possibly her own). Each subsequent person, arriving, sat in their own seat, if it was free, otherwise - in a random unoccupied one. What is the probability that the last person will sit in their own seat?
15. Complete the notebook shared with you via Moodle.

Problems for homework

16. A coin is flipped until one does not get a sequence "tails-tails-tails". What is the probability that the coin has been flipped
 - (a) exactly 6 times;
 - (b) exactly 7 times.
17. A cube with all of its faces painted is sawn into 343 smaller cubes of equal sizes. One of the small cubes is chosen at random. Find the probability that this cube:
 - (a) has exactly one painted face;
 - (b) has exactly two painted faces;
 - (c) has at least two painted faces.
18. List the elements of each of the following sample spaces:
 - (a) the set $S = \{x | 9 - 3x^2 \geq 0 \text{ and } x > 0\}$
 - (b) the set $S = \{x | 2x - 4 \geq 0 \text{ and } x < 1\}$.
19. m white balls and n black balls are placed at random in a row. Find the probability that k^{th} place is occupied with a black ball.
20. Six volumes are placed on a bookshelf in a random order. Find the probability that at least one of the volumes is not on its correct position.

21. Seventy numbers are chosen at random from integers $1, 2, 3, \dots, 100$. What is the probability that the largest number chosen is 98?
22. m white balls, n black balls and k yellow balls are placed at random in a row. Determine the probability that if looking at the balls from left to right, we will see a white ball before we see a black one?
23. Four people enter the lift at the ground floor of a nine-storey building. Find the probability that at least two of them get off at the same floor.
24. Six letters are chosen at random from the ones of the word PROPORTION. Determine the probability that it is possible to form the following words from these letters:
 - (a) OPTION;
 - (b) PORT;
 - (c) RIOT.
25. Twenty participants of a chess tournament are casting lots in order to get into one of two groups (each of the groups has 10 players in it). Find the probability that two of the strongest players are in the same group.
26. What is the probability to get two "threes" when rolling two dice given that the sum of the two numbers is a multiple of 3?
27. An urn contains 6 white balls, 4 black balls and 2 orange balls. Three balls are taken at random out of the urn. What is the probability that the balls are of the same colour?
28. Three dice are rolled. What is the probability that the product of all numbers rolled is 24?
29. It is known that some events A and B satisfy the equality $P(A) = P(B) = 0.5$. Find the difference $P(AB) - P(\overline{A}\overline{B})$
30. Find the largest and the smallest values of $P(ABC)$ if $P(A) = 0.7$, $P(B) = 0.8$, $P(C) = 0.9$. *Hint:* make sure that each value can be actually achieved.
31. It is known that $P(A) = \frac{2}{3}$, $P(B) = \frac{3}{4}$. Prove that $P(A|B) \geq \frac{5}{9}$.
32. Let A and B be independent events. Prove that
 - (a) A and $\overline{B}$ are independent;
 - (b) $\overline{A}$ and $\overline{B}$ are independent.
33. Two players are making turns in flipping a (symmetric) coin. The first of them to get tails wins the game. Find the probability for the first player to lose.
34. There are 7 white balls and 3 black balls in the first urn, 8 white balls and 4 black balls in the second urn, 2 white balls and 13 black balls in the third urn. One of the urns is chosen at random, and a random ball is taken out of it.
 - (a) Determine the probability that the ball is white.
 - (b) It is known that the ball taken out of the urn is white. What is the probability that the first urn was chosen out of the three?
35. A hospital specializes in curing three types of diseases: A , B and C . On average, there are 50% of patients who suffer from disease A , 30% of patients with disease B , and 20% of patients with disease C (each of patients has exactly one of these diseases). The probabilities to fully recover from the disease are equal to 0.95, 0.9 and 0.85 respectively. A patient who came to hospital recovered completely. What is the probability that he had disease B ?

36. The quantity of passenger cars that pass along the gas station is on average four times greater than that of trucks. The probability that a car refuels is equal to 0.05 for trucks and 0.15 for passenger cars. A car has just left the gas station. What is the probability that it is a truck?
37. Three marksmen are shooting at a target. The probabilities for them to hit the target are equal to $\frac{4}{5}$, $\frac{3}{4}$ and $\frac{2}{3}$ respectively. At least two hits are needed for the target to be destroyed. The marksmen fire one shot each, and the target is destroyed. What is the probability that the third marksman has hit the target?
38. Five people entered the lift at the ground floor of a nine-storey building. Find the probability that
- (a) none of them went out on fifth floor;
 - (b) (exactly) one of them went out at the fifth floor;
 - (c) (exactly) three of them when out at the fifth floor
39. What is more probable in case of two equally strong players: to win at least three games out of four or at least six out of eight? No draws are possible.
40. There are 10 devices installed on a workbench. During the trials each of them may break down with probability 0.15 independently from other devices. Find the probability that (exactly) three devices have broke down given that not all devices went through the trials successfully.
41. A marksman is shooting at a target until he misses three times. The probability to miss is equal to 0.2 for each of the shots he makes. What is the probability that
- (a) the marksman uses exactly 7 bullets;
 - (b) five bullets are sufficient for him?
42. Two players are playing the match (that consists of several games), each of the games can finish in favor of the younger player with probability 0.6 and in favor of the older player with the probability 0.4. The younger player has won exactly five games in the first eight games. What is the probability that he started the match with a defeat?

Answers

1. *Hint:* use the definition of conditional probability.
2. 0.5
3. 0.35
4. (a) $\frac{5}{64}$, (b) $\frac{9}{128}$
5. $\frac{5}{9}$
6. (a) 0.21; (b) 0.29; (c) 0.79
7. 0.98
8. **Answer:** (a) $(\frac{3}{4})^{10}$; (b) $(\frac{7}{8})^{10} - (\frac{6}{8})^{10}$; (c) $1 + (\frac{6}{8})^{10} - 2(\frac{7}{8})^{10}$; (d) $1 - (\frac{1}{8})^9$; (e) $1 - (\frac{1}{8})^{10} - (\frac{7}{8})^{10}$
9. $\frac{25}{48}$
10. (a) *Hint:* Use inclusion-exclusion; (b) *Hint:* apply the principle of mathematical induction
11. No. Yes.
12. $\frac{45}{91}$
13. ≈ 0.72
14. 0.5
16. (a) $\frac{1}{16}$, (b) $\frac{7}{128}$
17. (a) $\frac{150}{343}$, (b) $\frac{60}{343}$, (c) $\frac{68}{343}$
18. (a) $S = (0, \sqrt{3}]$, (b) $\emptyset$
19. $\frac{n}{m+n}$
20. $\frac{719}{720}$
21. $\frac{29}{462}$
22. $\frac{m}{m+n}$
23. $\frac{151}{256}$
24. (a) $\frac{1}{35}$, (b) $\frac{79}{210}$, (c) $\frac{5}{21}$
25. $\frac{9}{19}$
26. $\frac{1}{12}$
27. $\frac{6}{55}$
28. $\frac{5}{72}$
29. 0
30. *Hint:* use the results of Exercise 10 Maximum is 0.7, minimum is 0.4
31. *Hint:* use the results of Exercise 10
33. $\frac{1}{3}$

34. (a) 0.5; (b) $\frac{7}{15}$

35. $\frac{18}{61}$

36. $\frac{1}{13}$

37. 0.76

38. (a) $\left(\frac{7}{8}\right)^5 \approx 0.513$; (b) $5 \cdot \frac{1}{8} \cdot \left(\frac{7}{8}\right)^4 \approx 0.366$; (c) $10 \cdot \left(\frac{1}{8}\right)^3 \left(\frac{7}{8}\right)^2 \approx 0.015$

39. to win at least 3 games out of 4

40. $\frac{\binom{10}{3} \cdot 0.15^3 \cdot 0.85^7}{1 - 0.85^{10}} \approx 0.15$

41. (a) 0.049151; (b) 0.05792

42. $\frac{3}{8}$